MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION

Sidhant College of Engineering

A/P Sudumbre , Tal-Maval, Dist-Pune

Microproject

Academic Year 2026-27

Program Name:- Artificial Intelligence And Machine Learning **Scheme:-** K
Subjet Code:- Java Programming (314317) **Semester:-** 4th

A microproject on

Develop Employee Management System

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Under the guidence of

Miss.V.D.Patel

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CERTIFICATE

This is to certify that **Miss. Sakshi Diwane,** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in course java programming (314317) for the academic year 2025-2026 as prescribed in the curriculum.

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CERTIFICATE

This is to certify that **Miss. Anjum Mirza,** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in Course Java Programming (314317) for the academic year 2025-2026 as prescribed in the curriculum.

Place:-Sudumbare. Enrollment no:-25213380062

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This is to certify that **Miss. Dnyaneshwari Jambukar.** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in Course Java Programming (314317) for the academic year 2025-2026 as prescribed in the curriculum.

Place:-Sudumbare. Enrollment no:-24213380007

Date:- Seat no:-278627

Miss.V.D.Patel	**Mr.A.J.Gaikwad**	**Mr.M.N.Pohare**
Subject Teacher	H.O.D.	Diploma Co-ordinator

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1. Introduction to Employee Management System :

An Employee Management System (EMS) is a software application used to manage employee information in an organization. It helps companies store, organize, and update employee data easily.

The system keeps records such as employee name, ID, department, salary, attendance, and contact details. Instead of maintaining records manually on paper, the Employee Management System stores all information digitally in a database.

This system helps the organization to save time, reduce errors, and manage employees efficiently. It also makes it easier for managers to search, update, and maintain employee records.

Therefore, the Employee Management System is an important tool that helps organizations manage employee data in a simple and organized way.

2. Objectives of Employee Management System :

The main objectives of an Employee Management System are:

1. Maintain Employee Records :-

To store and manage all employee information such as name, ID, department, and salary.

2. Reduce Manual Work :-

To replace paper-based records with a digital system, making the work faster and easier.

3. Easy Data Access :-

To allow managers to quickly search and update employee details when needed.

4. Improve Accuracy :-

To reduce human errors in maintaining employee data.

5. Save Time and Effort :-

To help organizations manage employee information efficiently and quickly.

Employee Management :-

To organize employee records in a systematic and structured way.

3. Scope of the Project :

The Employee Management System project is designed to manage and organize employee information in an organization.

The scope of this project includes storing, updating, and managing employee details such as employee name, ID, department, salary, and contact information. The system helps administrators to add new employees, update existing records, and delete employee data when necessary.

It also allows quick searching and viewing of employee information, which makes employee management easier and faster. The project helps reduce manual paperwork and improves the efficiency and accuracy of managing employee records.

Therefore, the scope of this project is to provide a simple, efficient, and organized way to manage employee data in a computerized system.

4. Literature Review :

The Employee Management System has been studied and developed by many researchers and software developers to improve the management of employee information in organizations.

Earlier, organizations used manual systems such as paper files and registers to store employee records. This method was time-consuming, difficult to maintain, and prone to errors. Managing large amounts of employee data using manual methods also made searching and updating records very difficult.

With the development of computer technology and database systems, many software solutions were introduced to manage employee information efficiently. These systems allow organizations to store employee data in digital form, making it easier to add, update, delete, and retrieve information quickly.

Several studies have shown that using an Employee Management System can improve productivity, reduce paperwork, and increase data accuracy. Modern systems also include features such as attendance management, payroll management, and performance tracking.

Therefore, the literature review shows that the Employee Management System is an important tool for organizations to manage employee records efficiently and effectively using computerized systems.

5. Existing System :

In the existing system, employee information is usually managed using manual methods such as paper files, registers, and spreadsheets.

In this system, employee details like name, ID, department, salary, and contact information are recorded manually. Managing these records requires a lot of time and effort. Searching for specific employee information can also be slow and difficult.

The manual system may also lead to data duplication, errors, and loss of important records. It becomes more difficult to manage when the number of employees in the organization increases.

Therefore, the existing system is not very efficient, and it creates problems in managing employee records properly. This is why organizations need a computerized Employee Management System to improve efficiency and accuracy.

6. Limitations of Existing System :

The existing manual employee management system has several limitations.

1. Time Consuming :-

Maintaining employee records manually takes more time and effort.

2. High Chances of Errors :-

Manual data entry can lead to mistakes and incorrect information.

3. Difficult Data Search :-

Finding a specific employee record from many files or registers is slow and difficult.

4. Data Loss Risk :-

Important records may be lost or damaged if they are stored only on paper.

5. Large Paperwork :-

Manual systems require many files and documents, which need more storage space.

6. Poor Data Management :-

It becomes very difficult to manage and update records when the number of employees increases.

Therefore, the existing system is inefficient and needs improvement through a computerized Employee Management System.

7. Advantages of Proposed System :

The proposed Employee Management System (EMS) is a computerized system that makes managing employees much easier and more efficient compared to manual methods. Here's a detailed look at its advantages:

1. Saves Time :-

In a manual system, searching for an employee's details or updating records takes a lot of time. With a computerized system, you can find or update information in seconds, which saves a lot of effort for HR staff or managers.

2. Reduces Errors :-

Manual entry often leads to mistakes in data, like wrong salary, department, or contact details. The proposed system stores information digitally, which reduces errors and ensures data is accurate.

3. Easy Data Access :-

The system allows managers to search, view, and retrieve employee records quickly. You don't have to flip through files or folders; everything is available on the computer.

4. Efficient Record Management :

You can add new employees, update existing information, and remove old records easily. This makes employee management organized and systematic.

5. Secure Data Storage :-

All employee information is stored in a database, which is safer than paper records. It prevents data loss, damage, or misplacement, ensuring that important information is always available.

6. Can Handle Large Employee Data :-

As companies grow, the number of employees increases. The proposed system can handle hundreds or thousands of employee records without slowing down or becoming disorganized.

7. Improves Productivity :-

By automating routine tasks like attendance tracking, salary management, and record keeping, employees and managers can focus on more important work, improving overall productivity.

8. Long-Term Benefits :-

In the long run, the system helps the company save costs on manpower, reduce mistakes, and maintain accurate records for years. It also makes reporting, payroll, and performance evaluation.

The proposed Employee Management System is fast, accurate, secure, and efficient. It replaces the old manual system and provides a modern, reliable way to manage employee information, which benefits both the organization and its employees.

8. System Requirements for Employee Management System :

To successfully run the Employee Management System (EMS), certain hardware and software resources are required. These resources ensure the system runs smoothly, efficiently, and reliably.

1. Hardware Requirements :-

Hardware refers to the physical components of a computer needed to support the EMS.

Processor: Intel Core i3 or higher (for fast processing)

RAM: Minimum 4 GB (for smooth operation)

Hard Disk: At least 250 GB (for storing employee data)

Monitor: 15-inch or larger (for easy viewing)

Keyboard & Mouse: Standard input devices

Printer (Optional): For printing employee reports or records

Explanation:

The hardware ensures that the EMS can store data, process requests, and display information efficiently. A faster processor and enough RAM make the system quick and responsive.

2. Software Requirements :-

Software refers to the programs and applications needed to run and support the EMS.

Operating System: Windows 10 / Windows 11 / Linux

Database Software: MySQL, Oracle, or MS Access (for storing employee records)

Programming Language: Python, Java, or C# (to build the application)

IDE/Code Editor: Visual Studio, Eclipse, or PyCharm (for programming)

Web Server (Optional): Apache or XAMPP (if it's a web-based system)

Explanation:

The software provides a platform for coding, storing, and running the EMS. The database stores all employee details, while the programming language and IDE help design and implement the system.

The software requirements are the programs and applications needed to run the Employee Management System (EMS). These ensure the system works smoothly, securely, and efficiently.

1. Operating System (OS) :

Examples: Windows 10 / Windows 11 / Linux

Purpose: Provides a platform to run the EMS and manage hardware resources.

2. Database Software :

Examples: MySQL, Oracle, MS Access

Purpose: To store employee data such as employee ID, name, department, salary, and attendance.

3. Programming Language :

Examples: Python, Java, C#

Purpose: To develop the EMS application that interacts with the database and user interface.

4. IDE / Code Editor :

Examples: Visual Studio, Eclipse, PyCharm

Purpose: To write, edit, and debug the program code easily.

5. Web Server (Optional) :

Examples: XAMPP, Apache (if it is a web-based system)

Purpose: To host the application on a web interface, allowing access through browsers.

9. System Architecture of Employee Management System (EMS):

The System Architecture shows how the different parts of the Employee Management System work together. It explains how users, the application, and the database interact to manage employee data efficiently.

- Components of the System Architecture :-

a) User Interface (UI) :-

The part that the user interacts with.

Users (HR, managers, or admins) can add, update, delete, or view employee information.

b) Application Layer / Business Logic :-

Processes the data entered by the user.

Performs operations such as validating data, calculating salaries, and updating records.

c) Database Layer :-

Stores all employee data securely and systematically.

Stores details like employee ID, name, department, salary, attendance, and contact information.

- How the System Works :

1. The user inputs data through the User Interface.
2. The application layer processes the data according to the business logic.
3. The database layer stores or retrieves the data as needed.
4. The system displays the required .ion back to the user through the UI.

10. Data Flow Diagram (DFD) of Employee Management System (EMS) :

A Data Flow Diagram (DFD) shows how data moves through the Employee Management System. It helps us understand how information is processed, stored, and retrieved.

1. Level 0 DFD (Context Level) :

Shows the overall system and its interaction with the user.

Entities : User (HR/Admin).

Process : Employee Management System.

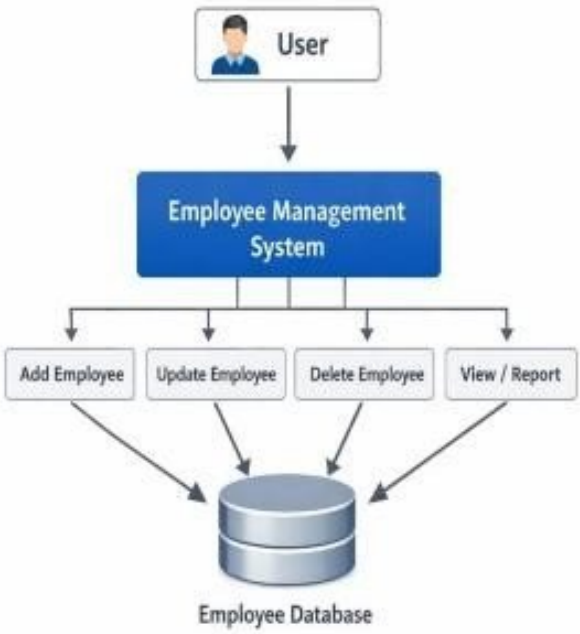
Data Store : Employee Database.

Explanation:The user sends requests like add employee, update details, or view information.

The system processes the request and interacts with the database.

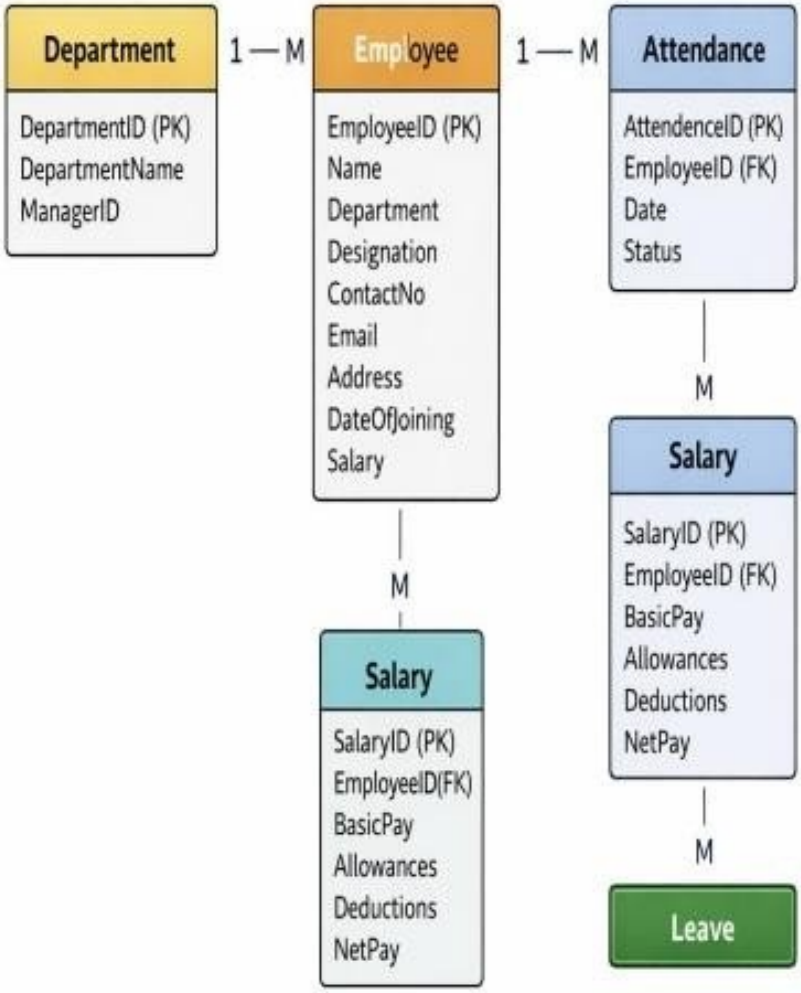
The database stores or provides data as required.

2. Level 1 DFD (Detailed Flow) :



Shows specific processes in the system. Processes include: Add Employee, Update Employee, Delete Employee, View Employee, Generate Report

ER Diagram Text Representation :



Explanation:

Each Employee is linked to a Department.

Employees have multiple Attendance, Salary, and Leave records.

Primary keys uniquely identify each record, while foreign keys connect related table

CODE:

```
import java.util.*

class Employee {
    int id;
    String name;
    double salary;

    // Constructor
    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }

    // Display method
    void display() {
        System.out.println(id + " " + name + " " + salary);
    }
}

public class EmployeeManagement {
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
ArrayList<Employee> list = new ArrayList<>();

int choice;

do {
    System.out.println("\n1.Add Employee");
    System.out.println("2.Display All");
    System.out.println("3.Search Employee");
    System.out.println("4.Exit");
    System.out.print("Enter choice: ");
    choice = sc.nextInt();

    switch(choice) {

        case 1:
            System.out.print("Enter ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Name: ");
            String name = sc.nextLine();
```

Output :

------ Employee Management System ---

1. Add Employee
2. Display All Employees
3. Search Employee
4. Exit

Enter your choice: 1

Enter ID: 101
Enter Name: Sakshi
Enter Salary: 25000
Employee Adrdes successfully!

1. Add Employee
2. Display All Employees
3. Search Employee
4. Exit

Enter your choice: 1

```
Enter ID: 102
Enter Name: Anjum
Enter Salary: 30000
Employee Added Successfully!


1. Add Employee
2. Display All Employees
3. Search Employee
4. Exit

Enter your choice: 3

Enter ID to search: 102

Employee Found:
102   Anjum   30000.0

1. Add Employee
2. Display All Employees
3. Search Employee
4. Exit
Enter your choice: 4

Exiting Program..
```

11. Conclusion :

The Employee Management System (EMS) is a computerized system designed to manage employee information efficiently. It replaces the traditional manual record-keeping, which is time-consuming and prone to errors, with a digital and organized system.

Key points:

The system stores employee data like ID, name, department, salary, attendance, and leave in a secure database.

It allows easy operations such as adding, updating, deleting, and viewing employee information.

The EMS improves accuracy, reduces manual errors, and saves time and effort for HR and management.

With testing and implementation, the system ensures reliable performance and is ready for real-world use

The system is scalable, allowing it to manage a large number of employees efficiently.

12. Reference :

1. Here is a simple and project-ready list of references you can include for your Employee Management System (EMS) Project:

2. Mathematics and Computer Science Books – For understanding database design and system logic.

3. Programming Books – Python, Java, or C# programming guides.

4. Database Books – MySQL, MS Access, or Oracle database tutorials.

5. Class Notes and Study Material – Notes provided by your teachers or professors.

6. Educational Websites –

7. W3Schools (https://www.w3schools.com□)

8. GeeksforGeeks (https://www.geeksforgeeks.org□)

9. TutorialsPoint (https://www.tutorialspoint.com□)

10. Online Research Papers – For Employee Management System concepts and implementation.

11. Library Books on Software Engineering and Database Management

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8. GeeksforGeeks (https://www.geeksforgeeks.org□)

9. TutorialsPoint (https://www.tutorialspoint.com□)

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